Linear screening and the dielectric function

Consider homogeneous electron system subject to an external (impurity) charge distribution $\rho_{ext}(\vec{r})$. The total charge of the impurity is $Q_{ext} = \int d^3\vec{r}\, \rho_{ext}(\vec{r})$. The electron charges of the conductor will redistribute themselves so as to <u>compensate</u> the external charge exactly $\Rightarrow$ $Q_{screening} = \int d^3\vec{r}\, \underbrace{\rho_s(\vec{r})}_{\text{density of mobile charges}} = -Q_{ext}$

The impurity + the screening charges produced screened potential

$$\phi(\vec{r}) = \int d^3\vec{r}\,' \frac{\rho_{ext}(\vec{r}\,') + \rho_s(\vec{r}\,')}{|\vec{r}-\vec{r}\,'|}$$

The classical theory of this is contained in Maxwell's eqn:

$$\vec{\nabla}\cdot\vec{E}(\vec{r}) = 4\pi\left(\rho_{ext}(\vec{r}) + \rho_s(\vec{r})\right) = 4\pi\,\rho_{total}(\vec{r}).$$

$$\vec{\nabla}\cdot\vec{D}(\vec{r}) = 4\pi\,\rho_{ext}(\vec{r})$$

We now F.T. these:

$$i\,\vec{q}\cdot\vec{E}(\vec{q}) = 4\pi\left(\rho_{ext}(\vec{q}) + \rho_s(\vec{q})\right)$$

$$i\,\vec{q}\cdot\vec{D}(\vec{q}) = 4\pi\,\rho_{ext}(\vec{q})$$

Thus we get <u>longitudinal</u> el. field $E_\ell(\vec{q}) = \dfrac{\vec{q}\cdot\vec{E}(\vec{q})}{q}$

and <u>longitudinal</u> displacement $D_\ell(\vec{q}) = \dfrac{\vec{q}\cdot\vec{D}(\vec{q})}{q}$

$$\Rightarrow \begin{cases} D_\ell(\vec{q}) = \dfrac{4\pi}{iq}\,\rho_{ext}(\vec{q}) \\ E_\ell(\vec{q}) = \dfrac{4\pi}{iq}\left(\rho_{ext}(\vec{q}) + \rho_s(\vec{q})\right) \end{cases}$$

$$\vec{E}_e(\vec{r}) = -\vec{\nabla}\phi(\vec{r}) \Rightarrow \vec{E}_e(\vec{q}) = -i\vec{q}\,\varphi(\vec{q})$$

$$\vec{q}\cdot\vec{E}_e(\vec{q}) = q\,\vec{E}_e(\vec{q}) = -i\,q^2\,\varphi(\vec{q})$$

$$\varphi(\vec{q}) = \frac{q\,\vec{E}_e(\vec{q})}{-i\,q^2} = \frac{i\,E_e(\vec{q})}{q}$$

The dielectric response function is (extending everything to ω-dependent)

$$\varepsilon(\vec{q},\omega) = \frac{D_e(\vec{q},\omega)}{E_e(\vec{q},\omega)} = \frac{\rho_{ext}(\vec{q},\omega)}{\rho_{ext}(\vec{q},\omega)+\rho_s(\vec{q},\omega)}$$

The essense of the <u>linear screening</u> is that for weak ρ_{ext}, $\rho_s \sim \rho_{ext}$ so that $\varepsilon(\vec{q},\omega)$ in fact becomes independent of the external charge distribution.

$\Rightarrow \varepsilon(\vec{q},\omega)$ is the property of the material.

Thus $$\phi(\vec{q},\omega) = \frac{i\,E_e(\vec{q},\omega)}{q} = \frac{4\pi}{q^2}\left(\rho_{ext}(\vec{q},\omega)+\rho_s(\vec{q},\omega)\right)$$

$$= \frac{4\pi}{q^2}\,\frac{\rho_{ext}(\vec{q},\omega)}{\varepsilon(\vec{q},\omega)}.$$

Frequently we write this as

$$\frac{1}{\varepsilon(\vec{q},\omega)} = \frac{\rho_{ext}+\rho_s}{\rho_{ext}} = 1 + \frac{\rho_s(\vec{q},\omega)}{\rho_{ext}(\vec{q},\omega)} = \frac{\rho_{TOT}(\vec{q},\omega)}{\rho_{ext}(\vec{q},\omega)}$$

We now consider response of electron gas to an applied external charge density $\rho_{ext}(\vec{r},t)$.

ρ_{ext} interacts with electrons via Coulomb interaction

$$\Rightarrow H'_{ext}(t) = \int d\vec{r} \int d\vec{r}' \; \rho(\vec{r})\, V(\vec{r}-\vec{r}')\, \rho_{ext}(\vec{r}',t)$$

electron charge density operator

time-dependent external density

Within linear response, the induced charge density

$$\langle \rho_{ind}(\vec{r},t)\rangle = +i \int_0^t dt' \, \langle [H'_{ext}(t'), \rho(\vec{r},t)]\rangle_{\hat{H}}$$

where the average and time-development are with respect to the t-independent hamiltonian $\hat{H}$ of the electron gas (which includes interaction between electrons!)

$$\langle \rho_{ind}(\vec{r},t)\rangle = -i \int_{-\infty}^{\infty} dt' \int d\vec{r}'\, d\vec{r}'' \, \langle [\rho(\vec{r},t), \rho(\vec{r}',t')\, V(\vec{r}'-\vec{r}'') \cdot \rho_{ext}(\vec{r}'',t')\rangle$$

$$= \int_{-\infty}^{\infty} dt' \int d\vec{r}' \int d\vec{r}'' \; K_R(\vec{r}-\vec{r}', t-t')\, V(\vec{r}'-\vec{r}'')\, \rho_{ext}(\vec{r}'',t')$$

$$K_R((\vec{r}-\vec{r}', t-t') = -i\,\theta(t-t') \langle [\rho(\vec{r},t), \rho(\vec{r}',t')]\rangle_{\hat{H}}$$

is retarded charge susceptibility. $\hat{H} = \hat{H}_0 + \hat{V}$

Kin. energy

Coulomb interaction

F.T. in space and time:

$$\langle \rho_{ind}(\vec{q},\omega)\rangle = V(\vec{q})\, K_R(\vec{q},\omega)\, \rho_{ext}(\vec{q},\omega)$$

Observe that $\rho_{ind}(\vec{q},\omega) = \rho_s(\vec{q},\omega)$ as screening charge density ρ_s appears in response to ρ_{ext}.

$$\Rightarrow \langle \rho_s(\vec{q},\omega)\rangle = V(\vec{q})\, K_R(\vec{q},\omega)\, \rho_{ext}(\vec{q},\omega)$$

$$\Rightarrow \frac{1}{\varepsilon(\vec{q},\omega)} = 1 + \frac{\rho_s(\vec{q},\omega)}{\rho_{ext}(\vec{q},\omega)} = 1 + V(\vec{q})\, K_R(\vec{q},\omega)$$

Dielectric function $\varepsilon(\vec{q},\omega)$ is determined by $K_R(\vec{q},\omega)$ which is calculated with full hamiltonian $\hat{H}$ of the system (but no H_{ext}!).

In case of <u>non-interacting</u> electron gas

$$\hat{H} \to \hat{H}_0 = \sum_k \xi_k\, a_k^+ a_k$$

and hence $K_R(\vec{q},\omega) \to \chi_0$ = charge susceptibility of Fermi gas

i.e. $$\chi_0(\vec{q},\omega) = \frac{1}{V}\sum_k \frac{n_{\vec{k}} - n_{\vec{k}+\vec{q}}}{\omega + i\delta - (\varepsilon_{\vec{k}+\vec{q}} - \varepsilon_{\vec{k}})}$$

which we calculated previously.

However full $K_R(\vec{q},\omega) \neq \chi_0(\vec{q},\omega)$

due to <u>interaction</u> between electrons

To calculate K_R for the <u>interacting</u> gas we consider its imaginary-time ~~[illegible]~~ cousin, $K_M(\vec{r},\tau)$, for which diagrammatic expansion does exist.

$$K_M(\vec{r},\tau) = -\langle \hat{T}_\tau \, \hat{\rho}(\vec{r},\tau)\, \hat{\rho}(0,0)\rangle = -\,\mathrm{Tr}\left(\left\langle \frac{e^{-\beta\hat{H}}}{Z}\, \hat{\rho}(r,\tau)\,\hat{\rho}(0,0)\right\rangle\right)$$

$$\hat{H} = \hat{H}_0 + \hat{V}.$$

Use interaction representation. Follow ch. 3 of Rickayzen (<u>pp. 62–71</u>) who describes this for single particle GF (Eqn. (3.13) – (3.40)). The same is described in ch. 3 of Mahan, "Many-particle physics," see ~~[illegible]~~ discussion (3.161) to (3.186).

Diagram rules are listed on <u>p. 135.</u>

$$K_M(\vec{r},\tau) = -\frac{\mathrm{Tr}\langle \hat{T}_\tau\, e^{-\beta\hat{H}_0}\, S(\beta)\, \hat{\rho}(\vec{r},\tau)\,\hat{\rho}(0,0)\rangle}{\mathrm{Tr}\left(e^{-\beta\hat{H}_0}\, S(\beta)\right)}$$

all operators here evolve with $\hat{H}_0$. $S(\beta)$ contains interactions.

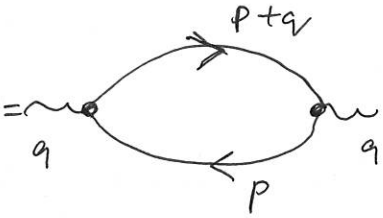
where $S(\beta) = \hat{T}_\tau \exp\left[-\int_0^\beta d\tau_1\, \hat{V}(\tau_1)\right] =$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \int_0^\beta d\tau_1 \ldots \int_0^\beta d\tau_n \left[\hat{T}_\tau\, \hat{V}(\tau_1)\ldots\hat{V}(\tau_n)\right]$$

is evolution operator in interaction representation. The denominator, $\mathrm{Tr}\left(e^{-\beta\hat{H}_0} S(\beta)\right)$, makes sure that we only need to consider <u>connected</u> diagrams.

$\underline{m=1}$ The first order diagram for K_M = polarization ~~operator~~ propagator

$$K_M^{(1)}(\vec{q}, i\omega_n) = \frac{2}{\beta V} \sum_{\vec{p}, ip_m} G^{(0)}(\vec{p}, ip_m)\, G^{(0)}(\vec{p}+\vec{q}, ip_m + i\omega_n)$$

$P^{(1)}(\vec{q},\omega) =$ 

Just a polarization bubble

$\rightarrow = G^{(0)}$ = electron GF

$\sim\sim = V(\vec{q}) = \frac{4\pi e^2}{q^2}$

$\underline{m=2}$

The whole series can be summed by introducing <u>irreducible</u> polarization propagator $P(\vec{q},\omega)$.
$P(\vec{q},\omega)$ is defined as a sum of diagrams which cannot be divided into two disconnected parts by cutting one interaction line ($\sim\sim$) only.

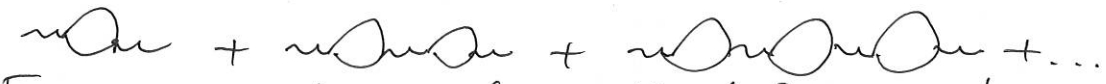
Then $K_M = P + PVP + PVPVP + \ldots$

$$= P + PV K_M$$

$$\Rightarrow K_M(1 - PV) = P \quad \Rightarrow \quad K_M = \frac{P}{1 - PV}$$ geometric series!

$$K_M(\vec{q},\omega) = \frac{P_M(\vec{q},\omega)}{1 - V_q P_M(\vec{q},\omega)}$$

The most diverging contribution is provided by RPA series where $P = P^{(1)}$



For $q \to 0$, each wavy line $V(q) = \frac{4\pi e^2}{q^2}$ diverges!

We now continue $K_M(i\omega_n) \to K_R(\omega)$

$$i\omega_n \to \omega + i\delta$$

$$K_R(\bar{q},\omega) = \frac{P_R^{(1)}(\bar{q},\omega)}{1 - V(\bar{q})\, P_R^{(1)}(\bar{q},\omega)}$$

Then $$\frac{1}{\varepsilon(\bar{q},\omega)} = 1 + V(\bar{q})\, K_R(\bar{q},\omega) =$$

$$= 1 + \frac{V(\bar{q})\, P_R^{(1)}(\bar{q},\omega)}{1 - V(\bar{q})\, P_R^{(1)}(\bar{q},\omega)} = \frac{1}{1 - V(\bar{q})\, P_R^{(1)}(\bar{q},\omega)}$$

$$\Rightarrow \varepsilon(\vec{q},\omega) = 1 - V(\vec{q})\, P_R^{(1)}(\vec{q},\omega) \quad !$$

[In our old notations, $P_R^{(1)}(\vec{q},\omega) = \chi_0(\vec{q},\omega)$]

<u>Consequences</u>

(1) Effective interaction is different from the bare one:

$$V_{eff}(\vec{q},\omega) = \frac{V(\vec{q})}{\varepsilon(\vec{q},\omega)}$$

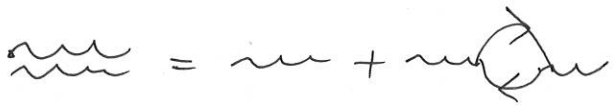
1) Screened
2) Frequency dependent

(2) Collective excitations when $\underline{\varepsilon(\vec{q},\omega) = 0}$:

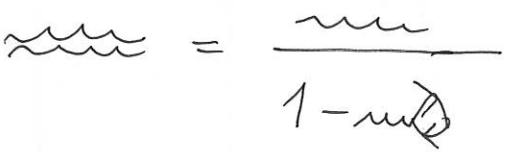
$$\rho_{TOT}(\vec{q},\omega) \cdot \varepsilon(\vec{q},\omega) = \rho_{ext}(\vec{q},\omega)$$

$\Rightarrow \rho_{TOT}$ can be non-zero even if $\rho_{ext} = 0$ when $\varepsilon(\bar{q},\omega) = 0$!

Interaction line $= V(\vec{q})$

Renormalized interaction 

$\Rightarrow$



Screening

$$\frac{\rho_s(\bar{q},\omega)}{\rho_{ext}(\bar{q},\omega)} = \frac{1}{\varepsilon(\bar{q},\omega)} - 1$$

Suppose external static ($\omega=0$) point charge $Q\,\delta(\vec{r}) \Rightarrow$
$\rho_{ext}(\vec{q},\omega=0) = Q$.

The screening charge is given by
$Q_s = \rho_s(\vec{q}=0,\omega=0)$ because $\int \rho_s(\vec{r})\,d^3\vec{r} = \rho_s(\vec{q}=0,\omega=0)$.

$$\Rightarrow \quad \frac{Q_s}{Q} = \frac{1}{\varepsilon(\vec{q}=0,\omega=0)} - 1 = \left(\frac{1}{1-V(q)\,P^{(1)}(\bar{q},\omega)} - 1\right)\Bigg|_{\substack{\omega=0\\ q=0}}$$

$$= \frac{V(\bar{q})\,P(\bar{q},\omega)}{1-V(\bar{q})\,P(\bar{q},\omega)}\Bigg|_{\substack{\omega=0\\ q=0}} = \frac{\frac{4\pi e^2}{q^2}\,P^{(1)}(\vec{q},\omega=0)}{1-\frac{4\pi e^2}{q^2}\,P^{(1)}(\vec{q},\omega=0)}\Bigg|_{q=0}$$

$$= \frac{P^{(1)}(\bar{q},\omega=0)}{\frac{q^2}{4\pi e^2} - P^{(1)}(\vec{q},\omega=0)}\Bigg|_{q=0} = \frac{P^{(1)}(\bar{q}=0,\omega=0)}{-P^{(1)}(\bar{q}=0,\omega=0)} = -1$$

provided that $P^{(1)}(\vec{q}=0,\omega=0)$ is finite.
And it is! $P^{(1)}(0,0) = -N_0 = -\nu$.

$\Rightarrow$ $Q_s = -Q$: the external charge is fully screened.

Screening: apply to Coulomb potential $V(\vec{q}) = \frac{4\pi e^2}{q^2}$

$$V_{eff}(\vec{q},\omega) = \frac{V(\vec{q})}{1 - V(\vec{q})\,\chi_0(\vec{q},\omega)}$$

Static response $\omega = 0$:

$$\chi_0(\vec{q},\omega=0) = -N(0)\ ,\quad N(0) = \text{density of states at the Fermi energy.}$$

$N(0)$ often denoted as ν.

$$\Rightarrow \chi_0(\vec{q}\to 0, \omega=0) = -\nu$$

$$\Rightarrow V_{eff}(\vec{q},0) = \frac{4\pi e^2/q^2}{1 - \frac{4\pi e^2}{q^2}(-\nu)} = \frac{4\pi e^2}{q^2 + 4\pi e^2 \nu} =$$

$$= \frac{4\pi e^2}{q^2 + \lambda_{TF}^{-2}}$$

Thomas-Fermi wavelength $\lambda_{TF} = \left(\frac{1}{4\pi e^2 \nu}\right)^{1/2}$

Potential is <u>screened</u>: while original $V(\vec{q}) = \frac{4\pi e^2}{q^2} \xrightarrow[q\to 0]{} \infty$

$$V_{eff}(\vec{q}\to 0, 0) \to \frac{1}{\nu} = \text{constant}$$

$$V_{eff}(\vec{r}) = \int \frac{d\vec{q}}{(2\pi)^3}\, V_{eff}(\vec{q})\, e^{-i\vec{q}\cdot\vec{r}} \sim \frac{e^{-\lambda_{TF} r}}{r}$$

Dynamic ($\omega \neq 0$) response

$\chi_0(\bar{q},\omega)$ is susceptibility of non-interacting Fermi gas.

Thus, interacting system responds with χ.
while non-interacting — with χ_0.

$\Rightarrow$ (1) Interactions renormalize response.

(2) One now has possibility of pole $1 - V(\bar{q})\chi_0(\bar{q},\omega) = 0$ which means that <u>collective excitations</u> are possible:

$$\left[1 - V(\bar{q})\chi_0(\bar{q},\omega)\right]\bar{\rho}(\bar{q},\omega) = \chi_0(\bar{q},\omega)\,\lambda(\bar{q},\omega):$$

If $\lambda(\bar{q},\omega) = 0$ we now have two options

(a) $\bar{\rho}(\bar{q},\omega) = 0 \rightarrow$ meaning, nothing happens.

(b) $\bar{\rho}(\bar{q},\omega_q) \neq 0$ because $V(\bar{q})\chi_0(\bar{q},\omega_q) = 1$

so that

$$\left[1 - V(\bar{q})\chi_0(\bar{q},\omega_q)\right]\bar{\rho}(\bar{q},\omega_q) = 0$$

is satisfied because expression in square brackets is <u>zero</u>

$1 = V(\bar{q})\chi_0(\bar{q},\omega_q)$ is implicit equation for ω_q = dispersion of collective mode.

Little thinking tells that these are actually two conditions:

$$\begin{cases} V(\bar{q})\ \mathrm{Re}\left[\chi_0(\bar{q},\omega_q)\right] = 1 \\ \mathrm{Im}\left[\chi_0(\bar{q},\omega_q)\right] = 0 . \end{cases}$$

<u>Plasmons</u>: consider $\omega \gg v_F q$ in $\chi_0(\bar{q},\omega)$.

Then $\mathrm{Im}\left[\chi_0(\bar{q},\omega)\right] = 0$ and

$$\mathrm{Re}\,\chi_0(\bar{q},\omega) = N_0 \left\{ -1 + \frac{\omega}{2v_F q} \ln \frac{\omega(1+v_F q/\omega)}{\omega(1-v_F q/\omega)} \right\} =$$

$X \equiv \frac{v_F q}{\omega} \ll 1$

$$= N_0 \left\{ -1 + \frac{1}{2x} \ln \frac{1+x}{1-x} \right\}$$

$$\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{1}{3}x^3 + \frac{1}{5}x^5 + \dots\right)$$

$$\Rightarrow \mathrm{Re}\,\chi_0 \cong N_0 \left\{ -1 + \frac{1}{2x}\left(2x + \frac{2}{3}x^3 + \frac{2}{5}x^5 + \dots\right) \right\}$$

$$= N_0 \left\{ -1 + 1 + \frac{1}{3}x^2 + \dots \right\} = \frac{1}{3} N_0 \left(\frac{v_F q}{\omega}\right)^2 .$$

Thus $V(\bar{q}) \cdot \frac{1}{3} N_0 \left(\frac{v_F q}{\omega_q}\right)^2 = 1$ is the condition.

See next page

$$\frac{4\pi e^2}{q^2} \frac{N_0}{3} \frac{v_F^2 q^2}{\omega_q^2} = 1 \Rightarrow \omega_q^2 = \frac{4\pi e^2}{3} N_0 v_F^2 .$$

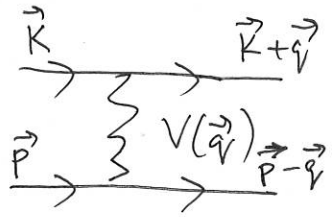
But $N_0 = \frac{m k_F}{2\pi^2}$, $v_F = \frac{k_F}{m} \Rightarrow \omega_q^2 = \frac{4\pi e^2}{3} \frac{k_F^3}{2\pi^2 m} = \frac{4\pi e^2 \rho_0}{m} =$ plasma frequency.

and $k_F^3 = 6\pi^2 \rho_0$, $\rho_0 = \frac{N}{V}$ = electron density.

Coulomb interaction.

Understanding renormalization of interaction <u>diagrammatically</u>

$$V_{eff}(\vec{q},\omega) = \frac{V(\vec{q})}{1 - V(\vec{q})\chi_0(\vec{q},\omega)}$$



$$\hat{H}_{int} = \frac{1}{2}\int d\vec{r}_1 d\vec{r}_2 \; V(\vec{r}_1-\vec{r}_2)\; \hat{\psi}^+(r_1)\,\hat{\psi}^+(r_2)\,\hat{\psi}(r_2)\,\hat{\psi}(r_1)$$

$$= \frac{1}{2}\int d\vec{r}_1 d\vec{r}_2 \; V(\vec{r}_1-\vec{r}_2)\; \frac{1}{V^2}\sum_{k_1k_2k_3k_4} e^{-i\vec{k}_1\vec{r}_1 - i\vec{k}_2\vec{r}_2 + i\vec{k}_3\vec{r}_2 + i\vec{k}_4\vec{r}_1}$$

$$\cdot\; a^+_{k_1} a^+_{k_2} a_{k_3} a_{k_4}$$

$$\vec{r}_1 - \vec{r}_2 = \vec{r}$$
$$\vec{r}_1 = \vec{R} + \tfrac{1}{2}\vec{r}$$
$$\vec{r}_2 = \vec{R} - \tfrac{1}{2}\vec{r}$$

$$= \frac{1}{2V}\int d\vec{r}\, d\vec{R}\; V(\vec{r})$$

$$\cdot\; \frac{1}{V}\sum_{k_1k_2k_3k_4} e^{-ik_1(R+\frac{r}{2}) - ik_2(\bar{R}-\frac{\bar{r}}{2}) + ik_3(R-\frac{r}{2}) + ik_4(R+\frac{1}{2}r)}$$

$$a^+_{k_1} a^+_{k_2} a_{k_3} a_{k_4}$$

$$\Rightarrow \quad \vec{k}_1 + \vec{k}_2 = \vec{k}_3 + \vec{k}_4 \quad \Rightarrow \quad \begin{array}{l} k_4 = k,\; k_3 = p, \\ k_1 = k+q,\; k_2 = p-q \end{array}$$

$$= \frac{1}{2V}\int d\vec{r} \sum_{k,p,q} a^+_{k+q}\, a^+_{p-q}\, a_p\, a_k\; V(\vec{r}).$$

$$\cdot\; e^{i\frac{\vec{r}}{2}\underbrace{(-\vec{k}-\vec{q}+\vec{p}-\vec{q}-\vec{p}+\vec{k})}_{-\vec{k}_1+\vec{k}_2-\vec{k}_3+\vec{k}_4}}$$

$$= \frac{1}{2V}\sum_{k,p,q} V(\vec{q})\; a^+_{k+q}\, a^+_{p-q}\, a_p\, a_k$$

$$V(\vec{q}) = \int d\vec{r}\; e^{-i\vec{r}\cdot\vec{q}}\; V(\vec{r}).$$

Note that for the Coulomb interaction

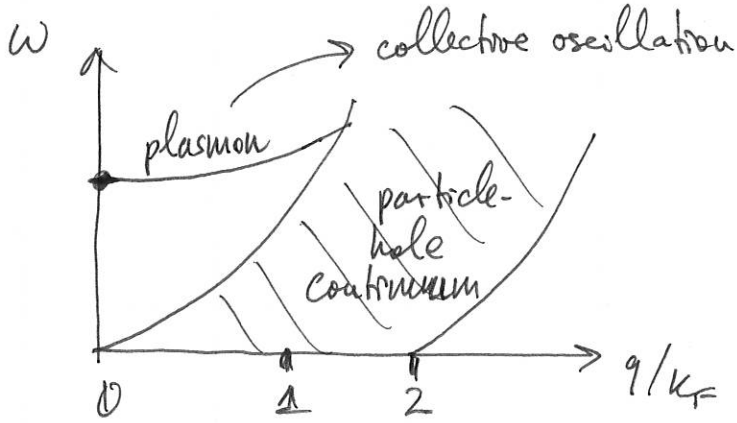
$$V(\vec{r}_1-\vec{r}_2) = V(\vec{r}) = \frac{e^2}{|\bar{r}_1-\bar{r}_2|} = \frac{e^2}{|\bar{r}|} = \frac{e^2}{r}$$

$$V(\bar{q}) = \int d\vec{r}\, V(\bar{r})\, e^{i\bar{q}\cdot\bar{r}} = \int_0^\infty dr\, r^2 \int_0^{2\pi} d\varphi \int_0^{\pi} d\theta \sin\theta \frac{e^2}{r} e^{iqr\cos\theta}$$

$$= 2\pi e^2 \int_0^\infty dr\, r \int_{-1}^{1} d(\cos\theta)\, e^{iqr\cos\theta} = (\xi = \cos\theta) =$$

$$= 2\pi e^2 \int_0^\infty dr\, r \left.\frac{e^{iqr\xi}}{iqr}\right|_{-1}^{1} = 2\pi e^2 \int_0^\infty dr \cdot r\, \frac{2i\sin(qr)}{iqr} =$$

$$= \frac{4\pi e^2}{q} \int_0^\infty dr\, \sin(qr) = \frac{4\pi e^2}{q^2} \underbrace{\int_0^\infty d(qr)\, \sin(qr)}_{-\cos(qr)\big|_0^\infty = +1.}$$

add regularization $e^{-\alpha r}$, and then set $\alpha = 0$ at the end.

$$V(\bar{q}) = \frac{4\pi e^2}{q^2}.$$